Depth Visualization and Overlay with Depth Anything

In this task, depth visualization and RGB-depth overlay were implemented using one frame from the **TUM RGB-D freiburg1_xyz video sequence**. The selected RGB frame was processed using **Depth Anything V2 Small** to generate a depth map.

The raw depth prediction was converted into a colorized depth image using the **PLASMA color map**. Then, the colorized depth image was combined with the original RGB image to create a depth overlay:

$$\text{Overlay} = (1-\alpha) \times \text{RGB} + \alpha \times \text{Depth} + \gamma$$

Different values of the parameter **alpha** were tested (0.2, 0.4, 0.5, 0.7, and 0.8) to analyze their effect on visualization.

Gamma (γ) add extra brightness/offset to the blended image which is not necessary in here so its value will be 0.

Effect of Alpha

The alpha value controls the contribution of the depth image in the final overlay.

- For lower alpha values, the RGB image remains dominant and the scene details are clearer, but depth information is less visible.
- For higher alpha values, the depth information becomes more dominant, but the original RGB appearance becomes less clear.
- Medium alpha values provide a balance between RGB details and depth information.

Selected Alpha Value

An interactive alpha slider was added to dynamically adjust the transparency of the colorized depth map and observe its effect on the interpretability of the RGB-depth overlay.

For user visualization, **alpha = 0.5** is selected because it provides a good balance between the original scene information and the depth representation. At this value, both object appearance and depth variations remain easy to interpret.

The experiment demonstrates that adjusting alpha allows controlling the trade-off between visual clarity of the original image and the visibility of depth information.

It is worth mentioning that if the goal is to identify and compare near and far objects, displaying the original RGB image and the colorized depth map side by side is preferable to combining them in an overlay. This approach preserves the details of the original image and facilitates the comparison and interpretation of the relative distances of objects.